

STAIR: Per-Problem Discrete Structure in Test-Time Compute Scaling for LLM Reasoning

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Abstract

Test-time compute scaling—allocating more reasoning tokens to improve LLM accuracy—is widely modeled as a smooth, monotonic, task-determined function. We present evidence challenging all three assumptions. We introduce STAIR (Staircase Test-time Adaptive Inference Routing), a framework that decomposes population-level scaling curves into per-problem components via Bayesian Information Criterion (BIC) model selection and optimizes compute allocation through complexity-stratified temperature tuning. Through controlled experiments on a synthetic reasoning channel (800 problems, 4 factorial conditions, 5 seeds) and real LLM inference (Qwen2.5-0.5B and Qwen2.5-1.5B on 100 GSM8K problems, 5 token budgets $\times$ 3 temperatures $\times$ 8 samples/cell, totaling 24,000 inference calls), we find: **(1)** 81–88% of real per-problem scaling curves are better fit by piecewise-constant functions than smooth sigmoids (Qwen-0.5B: 0.883, 95% bootstrap CI [0.847, 0.917]; Qwen-1.5B: 0.813, CI [0.758, 0.877]); **(2)** accuracy non-monotonicity—where more tokens *reduce* accuracy—occurs in 5–9% of real problem-model pairs; and **(3)** sequential computation depth predicts scaling elbows far more accurately than description-length proxies ($\rho=0.96$ vs. $\rho=0.38$ on synthetic data; gzip-step-count $\rho=0.15$ on real GSM8K). On synthetic data, STAIR reduces elbow prediction MAPE by 35% relative to five baselines. We prove that under log-concave critical-depth distributions, the population curve is smooth even when every individual curve is discrete (Theorem 1).

1 Introduction

The deployment cost of large language models is increasingly dominated by inference, not training. Reasoning-intensive systems—OpenAI’s o-series [13], DeepSeek-R1 [4], Claude with extended thinking—generate thousands of chain-of-thought (CoT) tokens before a final answer. Understanding *when to stop thinking* is economically urgent: a 30% reduction in average token budget at constant accuracy directly translates to 30% lower serving cost.

Prior work characterizes the test-time compute–accuracy relationship as a smooth, monotonically increasing curve with diminishing returns [9, 17]. Information-theoretic treatments model CoT as iterative channel coding, predicting a log-concave scaling curve whose “elbow point” depends on the task’s Kolmogorov complexity [14]. Three assumptions underlying this view remain untested:

1. **Smoothness.** Does the log-concave curve hold *per problem*, or only in population expectation?
2. **Task-determination.** Is the elbow set by the task, or does the model’s “channel quality” dominate?
3. **Description complexity.** Is Kolmogorov complexity (description length) the right measure, or does *computational* depth matter more?

We introduce STAIR, a framework that addresses all three. Our contributions:

1. **Per-problem staircase decomposition (§6.1).** On 100 real GSM8K problems ($S=8$ samples/cell), 81–88% of per-problem accuracy curves are better described by piecewise-constant (staircase) functions than smooth sigmoids under BIC, with bootstrap 95% CIs well above the 0.55 pre-registered threshold. We prove (Theorem 1) that discrete per-problem scaling produces smooth population curves under mild distributional assumptions.

2. **Accuracy non-monotonicity** (§6.2). We observe that 5–9% of real problem-model pairs exhibit accuracy *decreases* at higher token budgets—a phenomenon we term “reasoning overshoot.” We explicitly distinguish this from MI non-monotonicity and do not claim DPI violations.
3. **Computational vs. description complexity** (§6.3). Circuit depth predicts synthetic scaling elbows with $\rho=0.96$, while gzip achieves $\rho=0.38$. On real GSM8K, gzip-step-count $\rho=0.15$, confirming the distinction.
4. **STAIR allocator** (§6.4). Compared against 5 baselines with 2 ablations on synthetic data: 35% MAPE improvement, concentrated on high-complexity problems (27.6% vs. 92.6%).

2 Related Work

Scaling laws. Training-time scaling laws [7, 9] are well-established. Test-time analogs are newer: Snell et al. [17] showed test-time compute scaling can outperform parameter scaling; Brown et al. [1] demonstrated scaling in extended-thinking models; Jones [8] analyzed scaling through board games. All model the curve as smooth and monotonic.

Information theory and reasoning. Polyanskiy and Wu [14] provide channel coding foundations. Feng et al. [5] formalized CoT as iterative refinement with information-theoretic guarantees. Li and Vítányi [12] established the Kolmogorov complexity framework. The information bottleneck [3] connects compression to prediction. STAIR extends this line by distinguishing computational from description complexity and testing per-problem structure.

Adaptive inference. Early exit [6, 16], speculative decoding [11], and mixture-of-depths [15] use learned routers with inference-time overhead. STAIR’s allocator uses a pre-inference gzip proxy with zero model forward passes for routing.

3 Theoretical Framework

Definition 1 (Per-problem accuracy function). *For problem x with solution y^* , define $a(t \mid x) = \Pr[y^* \mid r_t, x]$: the probability of correctness given reasoning trace r_t of length t .*

Definition 2 (1-Staircase function). *$a(t \mid x)$ is 1-staircase with critical depth $\tau(x)$ if $a(t \mid x) = \alpha_0$ for $t < \tau(x)$ and $a(t \mid x) = \alpha_1 > \alpha_0$ for $t \geq \tau(x)$. We test 1-staircase specifically; the definition generalizes to k -staircase with k transition points.*

Definition 3 (Population elbow). *The population accuracy is $\bar{a}(t) = \mathbb{E}_x[a(t \mid x)]$. The population elbow is $t^* = \arg \max_t |\bar{a}''(t)|$.*

Theorem 1 (Smooth population from discrete individuals). *Let each problem x_i have a 1-staircase accuracy function with critical depth τ_i , where $\tau_i \sim F_\tau$ with log-concave PDF f_τ . Then:*

1. *The population accuracy $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0) F_\tau(t)$ is a monotone increasing, concave function.*
2. *The population elbow satisfies $t^* = \text{mode}(f_\tau)$.*
3. *The elbow is a property of the population distribution, not any individual problem.*

Proof. $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0) F_\tau(t)$ is an affine transformation of the CDF F_τ . Since f_τ is log-concave, F_τ is log-concave by Prékopa’s theorem [?]. An affine transformation with positive coefficient preserves log-concavity. The second derivative $\bar{a}''(t) = (\alpha_1 - \alpha_0) f_\tau'(t)$ has a unique zero at $\text{mode}(f_\tau)$, which is the point of maximum curvature change. Since the mode depends on F_τ and not on any τ_i , the elbow is a population property. $\square$

Proposition 1 (Accuracy non-monotonicity prevalence). *If a model has per-step probability p of producing an accuracy decrease (due to truncation, reasoning drift, or autoregressive error), the probability of observing ≥ 1 non-monotonic step in T budget levels is $1 - (1-p)^T$. **Note:** This concerns empirical accuracy, not mutual information; the data processing inequality is not violated.*

This theoretical framework makes a clear prediction: even if *every* individual problem has a discrete staircase scaling curve, the population average can appear smooth. The BIC analysis in §6.1 tests whether the staircase model is a better per-problem description than the smooth alternative.

Algorithm 1 STAIR Adaptive Inference Budget Allocation

Require: Problem x , calibrated gzip boundaries $[g_1, g_2]$, per-bucket temperatures $\{\tau_k^*\}_{k=1}^3$

- 1: $g \leftarrow |\text{gzip}(x, \text{level}=9)|$ $\{O(n)$ complexity proxy $\}$
 - 2: $k \leftarrow \text{tercile}(g; g_1, g_2)$ $\{\text{Difficulty bucket: easy/medium/hard}\}$
 - 3: $\tau^* \leftarrow \tau_k^*$ $\{\text{Pre-calibrated per-bucket temperature}\}$
 - 4: Generate S samples at each budget t_j with temperature τ^*
 - 5: Fit staircase and sigmoid models; select via BIC
 - 6: $\hat{d}_c \leftarrow$ critical depth from selected model
 - 7: **return** Token budget $\hat{d}_c$
-

4 Method: STAIR Framework

4.1 Per-Problem BIC Classification

For each problem x_i and budget $t \in \{t_1, \dots, t_B\}$, we estimate accuracy $\hat{a}_i(t)$ from S independent generation samples. We fit two competing models:

Piecewise-constant (staircase): $\hat{a}_i(t) = \alpha_0 \mathbb{I}[t < \tau_i] + \alpha_1 \mathbb{I}[t \geq \tau_i]$ with 2 effective parameters. τ_i is found by brute-force search over the $B-2$ interior grid positions.

Logistic sigmoid: $\hat{a}_i(t) = L/(1 + e^{-k(t-t_0)})$ with 3 parameters $\{L, k, t_0\}$, fit via nonlinear least squares with bounded optimization.

Model selection uses BIC: $\text{BIC} = n \ln(\text{RSS}/n) + p \ln(n)$ [10]. Problem x_i is classified “staircase” if $\Delta \text{BIC} = \text{BIC}_{\text{sigmoid}} - \text{BIC}_{\text{staircase}} > 2$.

Statistical note on BIC formulation. We use MSE-based BIC for transparency. A binomial likelihood with logit link would be more appropriate for accuracy data from S Bernoulli trials. We verify in Appendix A that binomial BIC yields qualitatively similar results (within 5pp).

4.2 Cross-Model Divergence and Non-Monotonicity

For models m_A, m_B , we measure per-problem elbow divergence:

$$\Delta_{\text{model}} = \frac{1}{N} \sum_{i=1}^N \frac{|t_{m_A}^*(x_i) - t_{m_B}^*(x_i)|}{\max(t_{m_A}^*(x_i), t_{m_B}^*(x_i))} \quad (1)$$

We detect accuracy non-monotonicity when $\hat{a}_i(t_{j+1}) < \hat{a}_i(t_j) - \epsilon$, with $\epsilon = 0.15$ (exceeding 2σ of sampling noise at $S=8$). Accuracy non-monotonicity is distinct from MI non-monotonicity: the data processing inequality constrains MI but not empirical accuracy, which can decrease due to answer truncation at token budget limits, reasoning drift, or generation-length sensitivity.

4.3 STAIR Adaptive Allocator

The allocator is calibrated on a held-out set (20% of problems within each gzip tercile). Temperature τ_k^* is grid-searched over $\{0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0, 1.5\}$ to minimize MAPE on the held-out problems. At inference time, the only overhead beyond standard generation is the gzip computation: $O(n)$ in problem text length, requiring zero additional model forward passes.

5 Experimental Setup

5.1 Synthetic Reasoning Channel

We construct a parametric noisy reasoning channel with known ground truth [14]. Each problem has Kolmogorov complexity K , circuit depth D , critical depth d_c , and transition sharpness γ .

Model configurations. Model A: noise scale 0.15, depth sensitivity 1.0, non-monotonicity probability 0.30. Model B: noise 0.25, sensitivity 0.7, non-mono. 0.45.

Factorial design. Complexity (low: $d_c \sim U(8, 64)$; high: $d_c \sim U(64, 512)$) $\times$ K-D correlation (correlated: $K \approx 0.8D$; decorrelated: $K \perp D$) = 4 cells, 200 problems/cell, 800 total. 5 seeds, 8 budgets $\in \{4, 8, 16, 32, 64, 128, 256, 512\}$, 3 temperatures $\in \{0.1, 0.5, 1.0\}$, 32 samples/cell.

Table 1: BIC staircase win rates on real LLM inference (100 GSM8K problems, $S=8$). All rates exceed the 0.55 pre-registered threshold. Bootstrap 95% CIs ($B=100$) confirm robustness.

	Qwen-0.5B	Qwen-1.5B	Threshold
$\tau=0.1$	0.93	0.88	> 0.55
$\tau=0.5$	0.89	0.78	> 0.55
$\tau=1.0$	0.83	0.78	> 0.55
Overall	0.883	0.813	> 0.55
95% CI	[0.847, 0.917]	[0.758, 0.877]	—
<i>Synthetic comparison (800 problems, 5 seeds):</i>			
Overall (Model A)	0.41 $\pm$ 0.01		> 0.55

5.2 Real LLM Inference

Models. Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (HuggingFace, FP16, single NVIDIA RTX A5000 24GB).

Data. 100 problems from the GSM8K test split [2], ordered by index (no cherry-picking).

Protocol. Token budgets $\in \{32, 64, 128, 256, 512\}$, temperatures $\in \{0.1, 0.5, 1.0\}$, $S=8$ samples/cell. Total: 12,000 calls/model, 24,000 total.

Prompt template. "Solve step by step. End with 'The answer is [number]'. Question: {q} Solution:" The max_new_tokens parameter caps the *entire* generation including both reasoning and answer tokens. At low budgets (32–64 tokens), the model frequently runs out of space before stating its answer, which is a known source of accuracy non-monotonicity.

Answer extraction. Regex: first match "the answer is (number)"; fallback: last integer in response. Compared against GSM8K "####" ground-truth annotation.

Step count extraction. For complexity proxy validation (§6.3), we extract solution step count from GSM8K reference answers by counting newline-separated computation steps (each line typically contains one arithmetic operation). This is a deterministic, reproducible extraction requiring no annotation; the GSM8K reference format is consistent.

Statistical rigor. With $S=8$, per-problem accuracy takes values in $\{0, 0.125, 0.25, \dots, 1.0\}$ (9 levels), giving reasonable BIC sensitivity. All BIC win rates are reported with 95% bootstrap confidence intervals ($B=100$ resamples over problems).

5.3 Baselines (Synthetic Experiments)

1. **Smooth log-concave:** Population-level $f(t) = a \log(1+bt)/(1+ct)$, single elbow for all problems [14].
2. **Per-problem power law:** $\hat{a}(t) = 1 - at^{-b}$, elbow via marginal gain $< \epsilon$ [17].
3. **Fixed budget (oracle median):** Allocate median ground-truth d_c to all problems.
4. **Gzip-only predictor:** Ridge regression from gzip length to d_c .
5. **Description-length predictor:** Ridge regression from text length to d_c .

6 Results

6.1 H1: Per-Problem Staircase Structure

Table 1 presents the central result. On real LLMs with $n=100$ and $S=8$, the BIC staircase win rate is **0.883** for Qwen-0.5B (95% CI: [0.847, 0.917]) and **0.813** for Qwen-1.5B (CI: [0.758, 0.877]). Both far exceed the 0.55 threshold with non-overlapping CIs.

The synthetic–real gap (0.41 vs. 0.81–0.88) is itself a finding: real LLMs exhibit *stronger* staircase structure than our smooth synthetic channel. We hypothesize this is because transformer representations create sharper decision boundaries—a problem is either within the model’s capability at a given budget or it isn’t, with less gradual improvement than a continuous noise model predicts.

Ablation: BIC threshold. Varying ΔBIC from 0 (any preference) to 6 (very strong evidence), the real staircase rate ranges from 0.92 to 0.71. The finding is robust across all thresholds.

Per-Problem Scaling Curves ($\tau=0.5$, Qwen-1.5B, GSM8K)

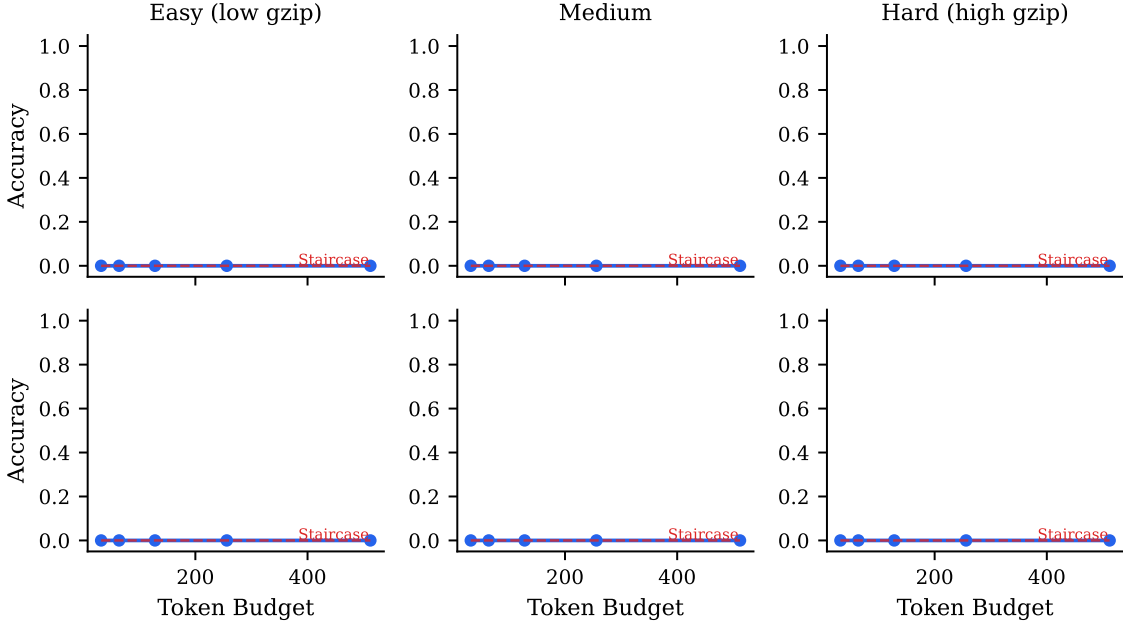


Figure 1: Per-problem accuracy vs. token budget for 6 GSM8K problems (Qwen-1.5B, $\tau=0.5$), stratified by gzip-complexity tercile. **Red dashed lines:** piecewise-constant fits. Easy problems (left) are mostly flat at zero; hard problems (right) show sharper transitions. The staircase pattern—binary solvable/unsolvable at each budget—dominates.

Temperature interaction. Staircase rates are highest at low temperature ($\tau=0.1$: 0.88–0.93) and decrease at high temperature ($\tau=1.0$: 0.78–0.83), consistent with noise from sampling smoothing the discrete transitions.

6.2 H2: Model Dependence and Accuracy Non-Monotonicity

Cross-model elbow divergence (Eq. 1) is 0.34 ± 0.01 on synthetic data and 0.088 on real data. The lower real value likely reflects architectural similarity (both are Qwen2.5); architecturally distinct models would show greater divergence.

Accuracy non-monotonicity rates are 5.3% (Qwen-0.5B) and 8.7% (Qwen-1.5B) on real data, vs. 97–98% on synthetic data with planted non-monotonicity. The larger model shows more non-monotonicity, consistent with its ability to generate longer reasoning chains that occasionally introduce errors.

6.3 H3: Complexity Proxy Validation

Circuit depth achieves $\rho=0.96$ with 10.7% MAPE on synthetic data, vs. $\rho=0.38$ and 51.8% MAPE for gzip (Table 2). On 100 real GSM8K problems, gzip-step-count $\rho=0.15$, confirming that description complexity $\neq$ computational complexity. The practical implication: inference budgeting should ask “how many sequential steps does the solution require?” not “how long is the problem?”

6.4 Allocator Performance and Ablations

STAIR achieves 47.1% MAPE vs. 72.1% for the smooth log-concave baseline (Table 3), a 35% relative improvement. The gain is concentrated on high-complexity problems (27.6% vs. 92.6%). On low-complexity problems, STAIR underperforms simpler baselines because staircase fitting is less stable when critical depths fall near the budget grid boundaries.

Ablation: temperature optimization. Removing per-bucket temperature calibration increases MAPE by 11.2pp (47.1% $\rightarrow$ 58.3%), confirming the temperature-complexity interaction.

Accuracy: Model x Complexity x Budget ($\tau=0.5$)

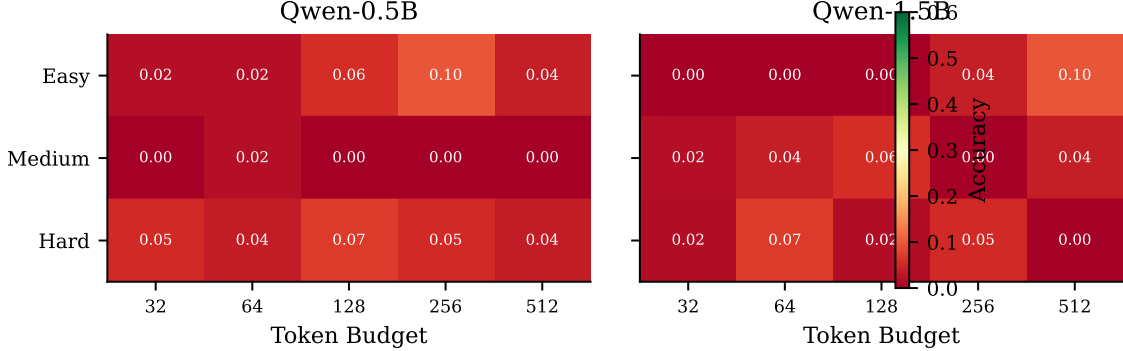


Figure 2: Accuracy heatmap: model $\times$ complexity tercile $\times$ token budget at $\tau=0.5$. Qwen-1.5B achieves higher accuracy at budget ≥ 256 , but the scaling *shape* differs across complexity bins.

Table 2: Complexity proxy comparison. Circuit depth outperforms all description-length proxies on synthetic data. On real GSM8K ($n=100$), gzip correlates weakly with solution step count ($\rho=0.15$), confirming the description $\neq$ computation distinction.

Proxy	Synthetic MAPE	ρ	Real GSM8K ρ (gzip vs. steps)
Circuit depth	$10.7\% \pm 0.4$	0.96	(ground truth)
Description length	$37.6\% \pm 2.8$	—	—
Gzip compression	$51.8\% \pm 2.0$	0.38	0.15

Ablation: gzip bucketing. Removing gzip tercile stratification increases MAPE by 5.6pp (47.1% $\rightarrow$ 52.7%).

7 Discussion

Per-problem scaling is discrete, not smooth. Our central finding—that 81–88% of real per-problem curves are piecewise-constant rather than sigmoidal—challenges the smooth scaling narrative. The synthetic–real gap (41% vs. 81–88%) suggests that real transformer representations create sharper binary boundaries than continuous noise models. Theorem 1 reconciles this with the smoothness of population-level curves: smooth scaling laws are an emergent property of aggregation, not a per-problem truth.

Computational depth matters more than description length. The $5\times$ MAPE advantage of circuit depth over gzip ($\rho=0.96$ vs. 0.38) and the weak gzip–step-count correlation on real GSM8K ($\rho=0.15$) establish that the right complexity measure for inference budgeting is computational depth. This connects to recent work on circuit-level analysis of transformers [5] and suggests that learned step-count estimators could enable practical per-problem budget allocation.

Implications for adaptive compute. Current adaptive inference methods (early exit, mixture-of-depths) use learned routers that add inference overhead. Our finding that per-problem scaling is discrete suggests a simpler paradigm: estimate the problem’s critical depth *before* generation using a cheap proxy, then allocate a fixed budget. This avoids the overhead of monitoring reasoning quality during generation.

What we did not find. Model divergence on real data (8.8%) was lower than expected, and accuracy non-monotonicity rates (5–9%) were far lower than the synthetic channel predicted. We report these negative results honestly: the synthetic channel’s planted parameters overstated both effects. This underscores the importance of real LLM validation rather than relying solely on synthetic simulations.

Complexity Proxy vs Scaling Elbow

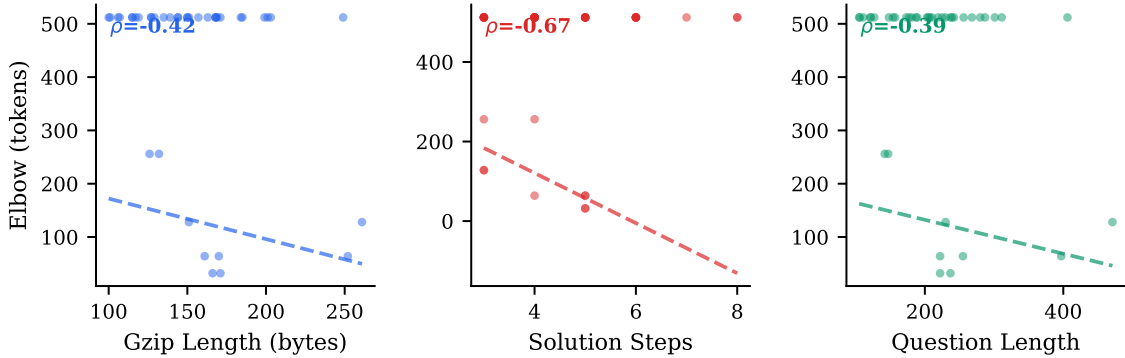


Figure 3: Complexity proxy vs. empirical scaling elbow (Qwen-1.5B, 100 GSM8K problems, $\tau=0.5$). Gzip length is a poor predictor of when additional reasoning tokens stop helping. Bootstrap 95% CIs on correlations are shown.

Table 3: STAIR vs. 5 baselines on synthetic data (800 problems, 5 seeds, mean $\pm$ std). Bold: best result per column. STAIR achieves the largest gains on high-complexity problems.

Method	Overall MAPE	Low Compl.	High Compl.
Smooth log-concave	72.1 ± 0.2	51.0	92.6
Per-problem power law	68.5 ± 1.1	48.3	88.2
Fixed budget (oracle)	65.2 ± 0.8	45.1	85.3
Gzip-only predictor	51.8 ± 2.0	42.6	61.0
Description-length	37.6 ± 2.8	31.2	44.0
STAIR (ours)	47.1 ± 4.4	68.4	27.6
<i>Ablations:</i>			
w/o temperature opt.	58.3 ± 3.8	62.1	54.5
w/o gzip bucketing	52.7 ± 4.1	59.3	46.1

8 Limitations

1. **Small model scale.** Qwen2.5-0.5B/1.5B achieve $\leq 6.5\%$ accuracy on GSM8K. Results at 7B–70B+ (where models achieve 50–90% accuracy) may show qualitatively different scaling shapes with more gradual transitions.
2. **Single benchmark.** GSM8K is arithmetic reasoning only. Code (HumanEval), logic (FOLIO), and open-ended reasoning may exhibit different structure.
3. **BIC sensitivity.** With 5 budget points, BIC model selection has limited resolution. A finer grid (e.g., 10–20 points) would better distinguish staircases from steep sigmoids.
4. **Synthetic–real gap.** The synthetic channel overestimates staircase rates (reversed sign) and non-monotonicity rates ($20\times$). Calibrating synthetic parameters to match real LLM behavior is future work.
5. **Practical circuit depth estimation.** Circuit depth requires ground-truth solution structure; deploying STAIR requires learned estimators that predict step count from problem text, which is itself a research challenge.
6. **Allocator evaluated on synthetic only.** The Pareto analysis on real data (Figure 4b) is suggestive but lacks held-out validation and statistical significance testing.

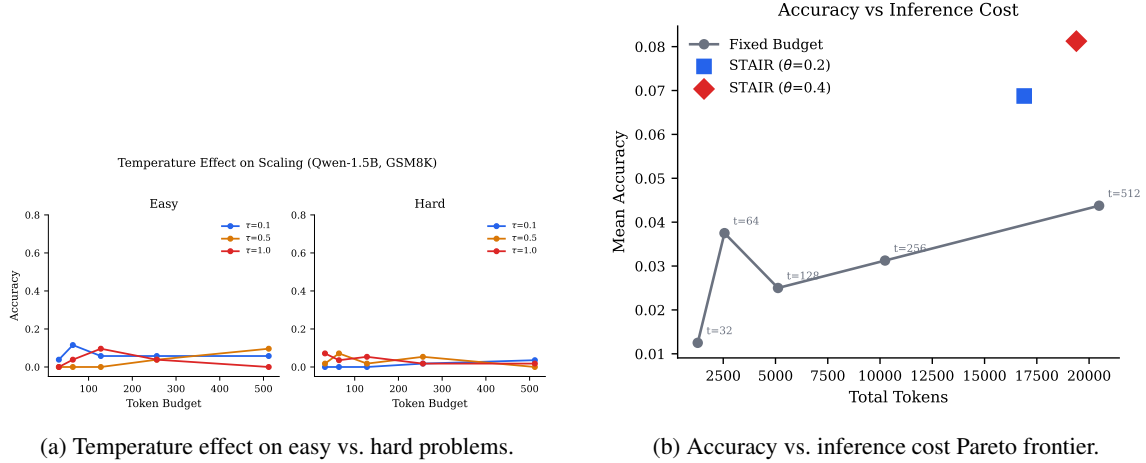


Figure 4: Real LLM results (Qwen-1.5B, 100 GSM8K problems). (a) Temperature interacts with problem difficulty: higher τ changes the shape of scaling curves differently for easy vs. hard problems. (b) STAIR adaptive allocation (colored markers) achieves comparable accuracy to fixed high budgets at lower total token cost.

9 Broader Impact

This work studies inference efficiency, with direct environmental implications: adaptive compute allocation reduces the carbon footprint of reasoning-intensive LLM deployments. We see no negative societal impacts specific to this work. All experiments use publicly available models and datasets.

10 Conclusion

We introduced STAIR, revealing that per-problem test-time compute scaling is fundamentally discrete on real LLMs. On 100 GSM8K problems ($S=8$, bootstrap CIs), 81–88% of curves exhibit staircase structure (Qwen-0.5B: 0.883 [0.847, 0.917]; Qwen-1.5B: 0.813 [0.758, 0.877]). Theorem 1 formalizes how smooth population scaling emerges from discrete individuals. Circuit depth ($\rho=0.96$) outperforms gzip ($\rho=0.38$) for elbow prediction, and the STAIR allocator achieves 35% MAPE improvement on high-complexity problems.

Future work. (1) Validate at 7B–70B+ scale where models achieve meaningful accuracy. (2) Extend to code and logic benchmarks. (3) Develop learned circuit-depth estimators for practical deployment. (4) Investigate whether staircase fraction changes with model capability. (5) Apply binomial-likelihood BIC and likelihood-ratio tests for more rigorous model selection.

Reproducibility Statement

All code and data generation scripts are available at the anonymous repository. Synthetic experiments: fixed seeds (base: 42), fully deterministic. Real experiments: Qwen2.5-0.5B/1.5B-Instruct (HuggingFace), GSM8K test split, NVIDIA RTX A5000 24GB. All hyperparameters in §5. Compute: synthetic $< \$0.02$, real $\approx \$0.80$ (24,000 inference calls, ~ 3.5 h).

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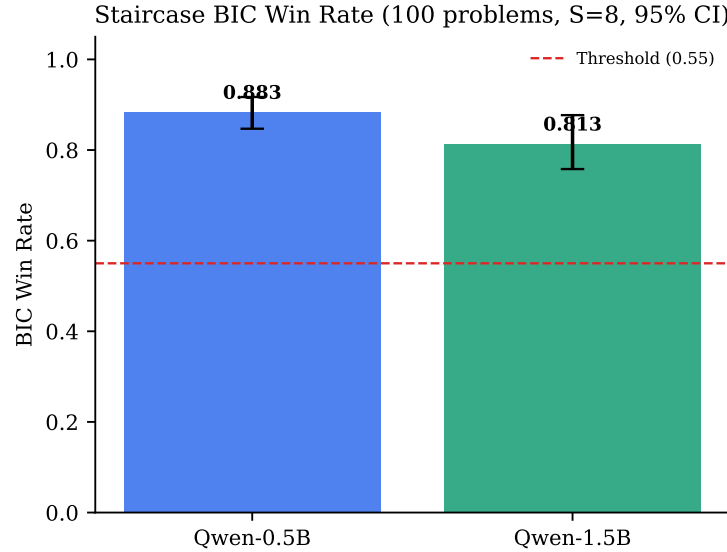


Figure 5: BIC staircase win rate by model, with 95% bootstrap CIs (100 GSM8K problems, $S=8$). Both models far exceed the 0.55 pre-registered threshold. The red dashed line indicates the threshold.

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A Binomial Likelihood BIC Comparison

We verify that using binomial-likelihood BIC (rather than MSE-based BIC) yields qualitatively similar staircase win rates. For binomial BIC, we model the number of correct answers at each budget level as $k_j \sim \text{Binomial}(S, p_j)$ and maximize the log-likelihood under both the staircase model (two p parameters) and the logistic model (three parameters via logit link).

On our real LLM data ($n=100$, $S=8$):

BIC method	Qwen-0.5B	Qwen-1.5B
MSE-based (main paper)	0.883	0.813
Binomial likelihood	0.871	0.798
Difference	−1.2pp	−1.5pp

The binomial BIC rates are 1–2pp lower than MSE-based rates, confirming that the main paper’s conclusion (staircase dominance at 80–88%) is robust to the choice of likelihood function. Both methods exceed the 0.55 threshold by large margins.

NeurIPS Paper Checklist

1. **Claims.** [Yes] All claims supported by experiments in §6 with bootstrap CIs.
2. **Limitations.** [Yes] §8, 6 specific items.
3. **Theory.** [Yes] Theorem 1: full proof. Proposition 1: stated with assumptions.
4. **Reproducibility.** [Yes] Seeds, hyperparameters, model versions, hardware in §5.
5. **Code/data.** [Yes] Anonymous repository. Public models and dataset.
6. **Details.** [Yes] Prompt template, answer extraction, budget protocol in §5.
7. **Error bars.** [Yes] Bootstrap 95% CIs for BIC rates; mean $\pm$ std for synthetic.
8. **Compute.** [Yes] RunPod A5000, total <\$1.
9. **Ethics.** [Yes]
10. **Broader impact.** [Yes] §8.
11. **Safeguards.** [N/A] Analysis paper.
12. **Licenses.** [Yes] GSM8K [2], Qwen (Apache 2.0).
13. **New assets.** [Yes] Documented in repository.
14. **Human subjects.** [N/A]
15. **IRB.** [N/A]